

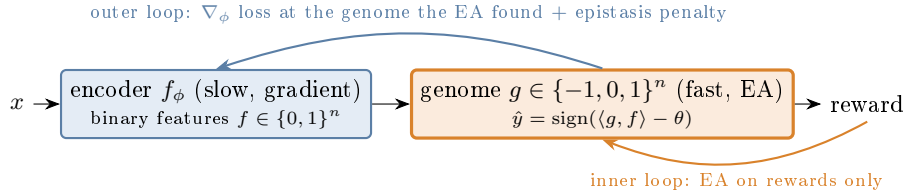
Evolvable Representations

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Learning slow features that make fast, gradient-free evolutionary adaptation provably easy

Question. Meta-learning shapes representations so that *gradient descent* adapts fast. What must a learned representation look like so that a simple *evolutionary algorithm*, seeing only scalar rewards, adapts fast, and provably so? Can such a representation be *trained*?

Why it matters. Deployed models mostly receive outcomes, not labelled gradients. Evolutionary algorithms (EAs) need only rewards, parallelise perfectly and handle discrete choices. They fail in 10^9 dimensions, but they are excellent on a tiny genome placed on top of a frozen, gradient-trained encoder, and then adaptation cannot cause forgetting. How fast they are depends on the *fitness landscape*, and that landscape is created by the representation.



Approach and first results on paper.

- **Definition.** The *evolvability* of a representation is the expected number of reward evaluations an EA needs to move between related tasks. It is driven by *epistasis* (ruggedness) and by the *task distance* k , the number of genome positions that must change between related tasks.
- **Theory.** On additive landscapes, a (1+1) EA re-adapts from distance k in $O(n \log k)$ evaluations: cost scales with the *size of the change*, not the model. New lemma: for threshold read-outs over binary features, $|\Delta_{ij}^2 F| \leq 2 \Pr[f_i = f_j = 1, |\langle g, f \rangle - \theta| < 4]$, so epistasis comes *only* from co-active features near the decision threshold. That is a feature property a training objective can control.
- **Method.** *Evolution-in-the-loop* meta-training: an inner EA solves tasks from rewards; the encoder is updated by first-order gradients at the EA's genomes, plus lemma-derived co-activation and margin penalties.

Plan.

- **Now (CPU cluster, one T4).** (1) Synthetic task family with known latent factors: measure epistasis, k and evaluations-to-adapt against ground truth. (2) Images with known factors (dSprites/Shapes3D). (3) Reward-only adaptation on CIFAR-100 rules.
- **Baselines:** multi-task, autoencoder and disentangled encoders; CMA-ES and policy gradients on continuous heads; ES-MAML; a labelled head as reference.
- **Go/no-go** after stage 1.
- **With compute:** the genome becomes per-expert routing biases in a mixture-of-experts LLM. Evolution-in-the-loop shapes experts and routers so that reward-only domain adaptation is fast. The prompt prefix is shared by the whole population, so N genomes cost roughly one forward pass.

Novelty. MAML, ANIL and ES-MAML shape an *initialisation* for a continuous learner. Evolvability ES optimises offspring diversity. LoraHub and C3PO search on top of a *fixed* library. None shapes the representation so that a discrete, reward-only search lands in a landscape class with runtime guarantees. The project connects EA runtime theory, meta-learning and the evolvability literature from biology (Wagner & Altenberg; Kashtan & Alon; Clune et al.).

Expected output. A measurable definition of evolvability for learned representations; a feature-level epistasis bound and re-adaptation theory beyond additivity; evolution-in-the-loop training; ground-truth benchmarks; and a scaling path to MoE routing. Full proposal (13 pp.) available on request.